Diachronic NLP Analysis of “Therapy-Speak” in General Web Discourse: ADHD and Autism in Common Crawl

Jakob Lütkemeier

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School of Social Sciences and Philosophy

Department of Political Science

Supervisor: Dr Tom Paskhalis

Course Directors: Dr Jeffrey Ziegler; Dr Tom Paskhalis

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Declaration

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Abstract

[Provisional] This dissertation investigates whether "therapy-speak" related to ADHD and autism has increased in general web discourse over the last decade, and how the surrounding language and framing may have shifted over time. Using Common Crawl as a large-scale web archive, I implement a reproducible pipeline that samples one crawl per year and extracts plaintext (WET) documents for trend estimation and a smaller, deeper corpus for downstream NLP analysis.

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1 Introduction

Reported rates of ADHD (Cui et al., 2026; McKechnie et al., 2023) and autism (Zeidan et al., 2022) have risen sharply. Concurrently, mental health awareness campaigns have proliferated (Bolinski et al., 2020), and mental health-related discourse has become increasingly common in everyday life (Foulkes & Andrews, 2023), giving rise to what has been described as *therapy-speak*. Therapy-speak refers to the imprecise and superficial integration of psychotherapy language into everyday communication (Isern-Mas & Almagro, 2025).

This concurrence has led some scholars to question the temporal relationship between rising prevalence rates and the increasing everyday use of mental health language. Foulkes and Andrews (2023) describe this as the *Prevalence Inflation Hypothesis*, which proposes a cyclical, intensifying relationship between the two phenomena. On one side of this cycle, rising prevalence rates may understandably increase mental health-related discourse in everyday life, hereafter referred to as therapy-speak. On the other side, the spread of therapy-speak may itself contribute to further increases in prevalence rates through overdiagnosis and overinterpretation (Foulkes & Andrews, 2023).

Before considering the potential risks of therapy-speak, its positive effects should be emphasised. The popularisation of therapy-speak may give people language for subtle emotional states (Medaris, 2024), improve mental health literacy (Schomerus et al., 2012), and thereby reduce stigma around mental illness (Fleary et al., 2022; Sampogna et al., 2017). It may also support better recognition and more accurate reporting of mental health problems (Foulkes & Andrews, 2023) and facilitate online identity and community formation, particularly around ADHD and autism (Ginapp et al., 2023).

At the same time, therapy-speak carries several risks that are predominantly epistemic in nature. These risks are not mutually exclusive and may overlap. First, therapy-speak may encourage overinterpretation, whereby milder and more transient forms of distress are conflated with mental health problems, often in conjunction with self-diagnosis (Foulkes & Andrews, 2023). At best, this may alter self-concept and behaviour (Alper et al., 2025; Foulkes & Andrews, 2023). At worst, when an individual interprets and labels their psychological experiences as a mental health problem, this may bring those symptoms into existence in the manner of a self-fulfilling prophecy. Adolescents may be particularly vulnerable to these dynamics because they are more prone to rumination, peer influence, and media messaging (Foulkes & Andrews, 2023).

Second, therapy-speak may contribute to the psychiatrisation of everyday suffering and distress (Brinkmann, 2014; Beeker et al., 2024). Third, mental health problems may be glamorised or romanticised, particularly among adolescents on social media (Ndour & Foulkes, 2025). Fourth, therapy-speak may spread inaccurate psychological information, including inaccurate information about autism (Aragon-Guevara et al., 2025) and ADHD (de Vries et al., 2025). Finally, therapy-speak may erode the meaning and relevance of mental-health-related terms, a process that can also be understood as trivialisation. This can contribute to hermeneutical injustice by depriving people who actually live with a certain mental condition of the words they need to describe their experiences (Isern-Mas & Almagro, 2025) and by reducing their symptoms to mere personality traits, thereby denying them a fully recognised psychiatric identity (Spencer & Carel, 2021).

This latter concern—the erosion of meaning in psychotherapy terms—is the focus of the present study, which investigates diachronic lexical semantic change in the terms ADHD and autism. These two terms stand out, arguably more than any others, when considering psychotherapy-related terms that have permeated mainstream discourse (Medaris, 2024). From January to May 2024, ADHD was the subject of 25,080 media articles, compared with 5775 articles during the equivalent period in 2014 (Martin et al., 2025). In May 2026, adhd and autism had been hashtagged in more than 5.2 million and 3.9 million videos on TikTok, respectively.

The following literature review first synthesises empirical evidence on lexical semantic change (LSC) in mental-health-related terms over recent decades, before reviewing the computational approaches used to study these processes and identifying the gap addressed by this project.

2 Literature Review

Lexical semantic change (LSC) refers to shifts in a word's meaning while its grammatical function remains stable, and constitutes a common form of language change (Campbell, 2013). For example, cloud, initially a meteorological term, broadened in usage to refer to internet-based data storage.

Concept Creep Theory (CCT) posits that harm-related concepts are particularly prone to LSC. Many harm-related terms, including addiction, bullying, harassment, prejudice, and trauma, appear to have expanded in meaning since at least the 1970s (Haslam, 2016; Haslam et al., 2020). Concept creep distinguishes between two forms of LSC that can occur concurrently: vertical creep and horizontal creep. Vertical creep refers to the loosening of definitions to include milder instances, whereas horizontal creep refers to the extension of definitions to encompass qualitatively new phenomena.

Previous studies have examined concept creep across different mental-health-related concepts. Vylomova and Haslam (2021) found that many of the terms they studied, including addiction, bullying, prejudice, harassment, and trauma, displayed both vertical creep and horizontal creep in a corpus of 825,628 psychology article abstracts from 875 journals and, to a lesser extent, in a general corpus combining the Corpus of Contemporary American English (CoCA) and the Corpus of Historical American English (CoHA). Baes, Vylomova, et al. (2023) found evidence of vertical creep in the word trauma in Vylomova and Haslam's psychology corpus. Baes, Haslam, and Vylomova (2023) similarly found vertical creep in mental-health-related concepts in the same psychology corpus, including addiction, anger, stress, and worry, as well as in Vylomova and Haslam's general corpus, including addiction, grief, stress, and worry.

Not all findings have aligned with this pattern. Xiao et al. (2023) found falling vertical creep for anxiety and depression in Vylomova and Haslam's psychology and general corpora, contrary to expectation. This unexpected result may reflect measurement unspecificity, particularly the absence of discourse context and construct distinctions. For example, depression may refer to psychiatric depression, but also to meteorological or economic phenomena. Similarly, anxiety may refer to a nosological category, a disorder construct, or an underlying human experience. Indeed, Pisl et al. (2025) showed that when psychiatric depression and anxiety as human experience were isolated, both displayed upward vertical creep. Iacob and Uban (2026) examined a range of common therapy-speak terms, including toxic, bipolar, psychopath, narcissistic, and triggered, and found mixed results. In an extension of Vylomova and Haslam's psychology corpus, these terms displayed horizontal creep. However, in a Reddit corpus comprising general Reddit comments and comments from psychology-oriented subreddits, results were mixed.

On balance, most mental-health-related concepts appear to have become milder and broader over the last few decades, although evidence for broadening is weaker.

Factors Influencing Concept Creep of Mental-Health-Related Terms

As noted above, general LSC is ubiquitous and naturally-occuring to some extent (Campbell, 2013). Frequency and polysemy are said to explain most variance in rates of lexical semantic change (Hamilton et al., 2016). However, the causes of the disproportionate levels of LSC observed in harm-related concepts remain uncertain.

Proposed explanations include cultural shifts toward greater sensitivity to harm, post-materialist values, and diminished exposure to adversity (Furedi, 2016; Haslam et al., 2020). In the case of mental-health-related terms, evidence is unclear as to whether the "creeping" of mental-health-related concepts reflects changes in official diagnostic criteria, broader sociocultural factors, or a combination of both.

Fabiano and Haslam (2020) showed that no revision of the Diagnostic and Statistical Manual of Mental Disorders (DSM) from the third edition onward was reliably more inflationary or deflationary overall. However, specific disorders changed significantly. Most notably, ADHD inflated by 18%, 33%, and then 17% in the three revisions following DSM-III. Rates of autism also inflated by 50% from DSM-III to DSM-III-R, albeit based on a single study, but deflated by 15% from DSM-IV to DSM-5.

2.1 Computational Approaches to Studying Diachronic Lexical Semantic Change

Since 2018, advances in deep learning have expanded the methodological repertoire for modelling semantic change (Manning, 2022). In particular, they have supported the development of language models that encode words as increasingly sophisticated vector representations. This marks a shift from count-based approaches, where word meaning is inferred from patterns of co-occurrence, toward prediction-based embeddings, where vectors are learned iteratively through a language-modelling objective (Grimmer et al., 2022; Jurafsky & Martin, 2026).

Building on these methodological developments, Baes et al. (2024) proposed the *SIBling* (**S**entiment, **I**ntensity, and **B**readth) framework as a unified and multidimensional approach to LSC. The framework situates CCT within a broader account of LSC and links this conceptual model to computational methods, including both count-based and embedding-based techniques. It distinguishes at least three dimensions along which terms may change over time: vertical drift, horizontal drift, and sentiment.

Vertical drift refers to changes in severity or intensity. A term may become stronger, as in hilarious shifting from cheerful or amusing to extremely funny, or weaker, as in trauma shifting from brain injuries to milder events such as business loss. This dimension maps onto vertical concept creep. Horizontal drift refers to changes in breadth. A term may become narrower, as in doctor shifting from scholar or teacher to primarily denoting a medical professional, or wider, as in cloud shifting from a meteorological term to internet-based data storage. This dimension maps onto horizontal concept creep. Sentiment refers to changes in connotation. A term may acquire a more positive connotation, as in geek shifting from a derogatory term for odd people to someone passionate about a field, or a more negative connotation, as in retarded shifting from a neutral term for intellectual disability to a highly pejorative term. This dimension is roughly equivalent to destigmatisation and stigmatisation.

According to SIBling, these dimensions can be complemented by evaluating shifts in the frequency of target words and in the thematic content of their collocates.

2.2 Empirical Gap

Two gaps motivate the present study. The first concerns the evidentiary basis of existing work. Research on lexical semantic change in mental-health-related concepts has relied heavily on curated—and often the same—corpora. Studies of general language have used Vylomova and Haslam (2021) combined CoCA and CoHA corpus (Baes, Haslam, & Vylomova, 2023; Baes et al., 2024; Xiao et al., 2023), while studies of psychology-specific language have repeatedly drawn on Vylomova and Haslam (2021) psychology corpus (Baes, Haslam, & Vylomova, 2023; Baes et al., 2024; Iacob & Uban, 2026; Xiao et al., 2023). Other analyses have turned to Reddit data (Iacob & Uban, 2026; Kang et al., 2025).

These corpus choices are consequential because the broader objective is to infer cultural dynamics from lexical change. Curated corpora may reflect shifts in editorial policy, audience orientation, or ideological stance rather than wider changes in public discourse (Pisl et al., 2025). Reddit, similarly, is not a neutral proxy for general discourse, as its user base, like that of many social media platforms, is demographically skewed (Gjurković et al., 2021). To date, research on lexical semantic change in mental-health-related concepts has not examined general public web discourse.

The second gap concerns the target concepts themselves. Direct evidence on lexical semantic change in ADHD and autism remains sparse, although Kang et al. (2025) showed that ADHD and autism appear to have converged on Reddit. From 2019 onward, their semantic similarity increased, with ADHD and autism becoming more contextually similar than comparison conditions.

The present study addresses these two gaps.

2.3 The Current Study

The present study is guided by three research questions: (i) how has the prevalence of ADHD- and autism-related terminology in general web discourse changed from 2014 to 2026; (ii) how have the semantic profiles of ADHD and autism evolved over this period, relative to non-clinical negative baseline emotion terms; and (iii) how has the thematic framing of ADHD and autism changed over time in public web discourse?

The web data are collected using Common Crawl. We built an economical and reproducible pipeline to extract quality-gated and substantive web content around ADHD and autism, as well as baseline terms.

Methodologically, the study adopts Baes et al. (2024) SIBling Framework with two deliberate deviations. First, for measuring intensity and sentiment, the study uses Mohammad (2025) NRC-VAD Lexicon rather than Warriner norms. This dictionary contains more terms, 20,007 compared with 13,915, and has been shown to be substantially more reliable, with reliability of 0.923 compared with 0.823 aggregated across three dimensions (Mohammad, 2025). Second, instead of SIBling's top-down, dictionary-based thematic index geared toward measuring pathologisation, the study employs a bottom-up topic-modelling approach to capture emergent framings of the target terms, such as clinical versus identity/neurodiversity narratives, education/workplace accommodation, and social-justice language.

In all, this project contributes by extending concept creep and therapy-speak research to ADHD and autism; shifting evidence from platform-specific (Iacob & Uban, 2026; Kang et al., 2025) or curated corpora (Baes, Haslam, & Vylomova, 2023; Baes, Vylomova, et al., 2023; Baes et al., 2024; Xiao et al., 2023) toward general web discourse; implementing and making available an economical, reproducible Common Crawl pipeline designed to extract high-quality general discourse to study diachronic lexical-semantic change in target terms against matched baseline terms; and validating and extending the SIBling framework.

Consistent with the gravity of the literature (Baes, Haslam, & Vylomova, 2023; Baes, Vylomova, et al., 2023; Iacob & Uban, 2026; Vylomova & Haslam, 2021; Xiao et al., 2023), we hypothesise that ADHD and autism have become more frequent and their collocates milder and broader since 2014. Given the absence of prior findings, the present study adopts a conservative approach and proposes no direction regarding changes in sentiment or thematic content among terms collocating with the target terms.

3 | Materials

3.1 Common Crawl

Common Crawl is the largest freely available public archive of web crawl data and is released in recurring crawl snapshots (Baack, 2024; Common Crawl Team, 2024). It is widely used as a web-scale source of language data for search, corpus construction, NLP research, and large language model pretraining (Baack, 2024; Wenzek et al., 2019). For this project, its central value lies in breadth: Common Crawl provides repeated cross-sections of general web discourse rather than data from a single platform, newspaper archive, or curated clinical corpus. However, it should not be treated as a representative survey of the web or of public opinion. The resulting data reflect what Common Crawl crawled, retained, and made available.

To collect the data for this study, we built a Common Crawl collection pipeline designed to extract high-quality general discourse for the analysis of diachronic lexical-semantic change in specific target terms, here ADHD and autism, against matched baseline terms. Target and baseline terms are processed together so that yearly denominators, sampling logic, and quality filters remain comparable across term groups. The pipeline uses Common Crawl's two main file formats sequentially. WET files provide compact extracted plaintext and are therefore used for large-scale term scanning and yearly prevalence denominators, but they lack the HTML structure and metadata needed for stronger validation. Candidate documents are therefore resolved to their corresponding WARC records, which preserve the archived web response, including HTML and headers. Although WARC processing is slower, it enables main-text extraction, term-survival validation, and metadata recovery. This WET-first, WARC-second design keeps the pipeline economical by reserving expensive WARC processing for candidate documents only.

The design consists of two linked tracks. The trend track uses fixed-effort annual samples to estimate how frequently target and baseline terms appear over time. The corpus track builds a larger, quality-gated document corpus for downstream NLP analysis. This two-track structure separates rate estimation from corpus construction: the trend track facilitates comparability across years, whereas the corpus track prioritises document quality and sufficient target-term coverage. To reduce the risk that a small number of large websites dominate the corpus, domain caps are applied at 50 WET-validated hit rows per registered domain per Common Crawl crawl. Intermediate summaries and manifests are retained so that each year, crawl, track, and batch remains auditable. The pipeline is built to run on AWS infrastructure, using S3 to store intermediate artefacts and EC2 for network-bound Common Crawl index lookup and WARC byte-range extraction. Yearly crawl selection is deterministic: one Common Crawl snapshot is selected per year near a fixed annual anchor date and then frozen in a crawl map.

The collection spans 13 annual Common Crawl snapshots from 2014 to 2026. In the trend track, the pipeline scanned 27.7 million WET records [provisional] and retained 78,899 WARC-validated term hits [provisional]. In the corpus track, it scanned 28.1 million WET records [provisional] and retained 42,975 analysis-ready documents [provisional]. Target-term coverage comprises 10,998 target documents [provisional], including 3,907 ADHD documents [provisional] and 8,558 autism documents [provisional]. The collection was run on an AWS `m7i-flex.large` instance, featuring 2 vCPUs, 8 GiB of RAM, and up to 12.5 Gbps network bandwidth (AWS, n.d.). On this instance type, corpus throughput was approximately one hour per million scanned WET records [provisional].

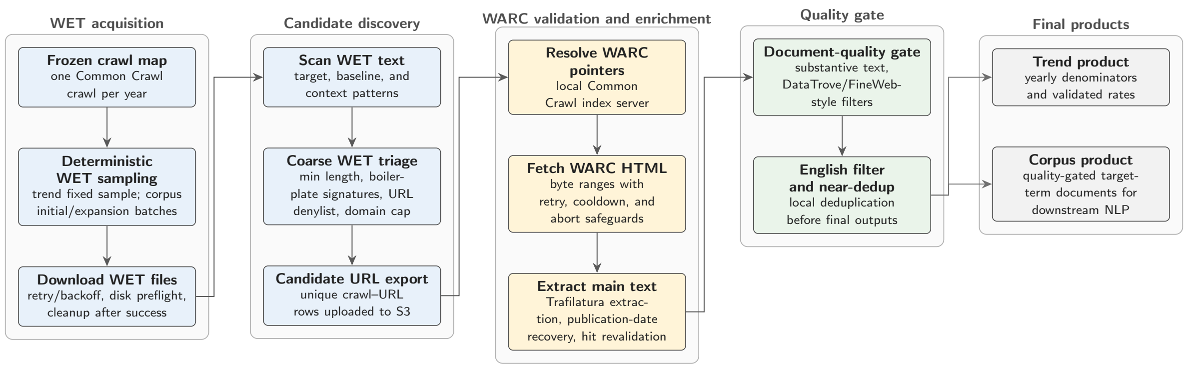

Figure 3.1: Common Crawl collection pipeline used to construct the trend and corpus materials.

The collection workflow is illustrated in Figure 3.1. It proceeds as follows. First, one Common Crawl snapshot is selected for each year, and WET files are sampled deterministically. Second, WET plaintext is scanned for target and baseline term matches, with document counts retained as the trend denominator. Third, conservative WET-stage triage is applied to remove obvious page chrome, directory pages, search or listing pages, product or job pages, and spam-like URL patterns. Fourth, unique candidate URLs are exported and resolved to WARC filenames, offsets, and byte ranges using Common Crawl index metadata. Fifth, the corresponding WARC records are fetched, and main text is re-extracted from the archived HTML. Sixth, documents are retained only when the matched terms survive WARC-based main-text extraction using Trafilatura version 2.0.0 (Barbaresi, 2021). Seventh, documents are enriched with candidate publication dates where available, while crawl capture year is treated as the primary temporal anchor. Eighth, post-WARC document-quality filters are applied to remove non-substantive or structurally unsuitable pages. Ninth, English documents are retained and near-identical texts are deduplicated within the collection. Finally, processed outputs are written separately for the trend track and corpus track. The collection pipeline is available in the project repository.[1]

Preprocessing

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3.2 Target Terms

Two target concepts were selected for diachronic lexical semantic change analysis: *ADHD* and *autism*. Target documents were retrieved using the matching expressions shown in Table 3.1. The abbreviation `ASD` was retained only when *autism* occurred within ±200 characters, reducing false positives from unrelated acronym use.

Table 3.1: Target-term matching expressions.

Concept	Matching expressions
ADHD	`\badhd\b`; `attention[-\s]?deficit`
Autism	`\bautism\b`; `\bautistic\b`; `autism[-\s]?spectrum`; `\bASD\b`

For comparison, I selected three negative, non-clinical emotion terms with sufficient coverage and interpretable usage: *frustration*, *sadness*, and *loneliness*. These terms are not exact semantic controls for ADHD and autism; they provide a baseline for separating target-specific change from broader shifts in negative affective language in the corpus.

3.3 NRC–VAD Lexicon

The affective analyses use the NRC Valence, Arousal, and Dominance (VAD) Lexicon v2.1 (Mohammad, 2025). The lexicon provides real-valued ratings for approximately

[1] https://github.com/jako6f/msc-nlp-therapy-speak

55,000 English words and phrases. Scores range from 0 to 1, where higher values indicate greater valence, arousal, or dominance.

This study uses valence to estimate whether target-term contexts become more positive or negative over time, and arousal to estimate changes in emotional intensity. Dominance is present in the source lexicon but is not used. The ratings were produced through Best–Worst Scaling, where annotators are given four items (4-tuple) and asked which item is the Best (highest in terms of the property of interest) and which is the Worst (least in terms of the property of interest).

4 Method

4.1 Salience (Prevalence)

Using the Trend Track, yearly mention rates will be estimated as relative frequencies (candidate and validated hits per scanned corpus size), enabling a diachronic test of whether these terms are becoming more prominent in general web discourse. *[status: incomplete]*

4.2 Intensity (Vertical Creep)

Collocates within a ±5-word window of the target term will be extracted per year and scored using the NRC Valence–Arousal–Dominance lexicon (M. Saif, 2025). Annual weighted arousal indices (and derived "severity" composites, where appropriate) will test whether the surrounding language becomes more or less emotionally intense over time. *[status: incomplete]*

4.3 Breadth (Horizontal Creep)

Breadth will be estimated via contextual dispersion: a fixed number of sentences containing the target term will be sampled per year, embedded with transformer sentence embeddings, and summarised by average pairwise cosine distance. Increasing dispersion indicates a widening range of contexts (semantic broadening). *[status: incomplete]*

4.4 Sentiment (Connotation)

Using the same collocate windows, annual weighted valence indices (NRC VAD) will test for shifts toward more positive vs more negative connotations (amelioration vs degeneration), reflecting potential destigmatisation or increased pejoration. *[status: incomplete]*

4.5 Thematic Evolution

The Corpus Track will be analysed with BERTopic (transformer embeddings, UMAP + HDBSCAN clustering, c-TF-IDF representations) to trace topics-over-time in annual bins. *[status: incomplete]*

4.6 Analytic Strategy

Linear regression analyses were performed to test the statistical significance of the predicted trends in the hypotheses....

5 Results

6 | Discussion

7 | Conclusion

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Appendices